

Bart Gajderowicz, PhD

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Address: Toronto, Ontario

Citizenship: Canadian, Polish

Professional Summary

I have advanced experience in software development and knowledge modelling, focusing on simulation, reasoning, machine learning, data analytics, and system architectures. I am highly skilled at problem-solving with analytical reasoning. I enjoy teaching through student engagement and experiential learning methodologies. I have extensive experience in outreach, collaboration, and bridging the gap across different fields of study and application domains. My research focuses on modelling processes, organizational behavior, and computational sociology, with applications in smart cities and social service provisioning.

Research Fields

Artificial Intelligence, distributed systems, distributed problem decomposition, event-driven microservices, organizational behavior, knowledge representation (symbolic), machine learning (sub-symbolic), hybrid-AI (neural-symbolic), computational sociology, social simulation, AI planning, multi-agent systems, ontology engineering, knowledge graphs, software engineering, human decision-making augmentation, cognitive science, and human-system interactions in complex system-of-systems paradigm.

Technical Skills

Main Languages: Python, NetworkX, Prolog, OWL 2, RDF, Ruby, SPARQL, SQL

Main Methodologies: Ontologies (reasoning with OWL 2, RDF, SWI-Prolog, engineering with Protégé), Databases (relational DBMS, graph-based DBMS, knowledge graphs), simulation (agent-based modelling, GAMA), Machine Learning (BERT, Pytorch, Sci-Kit Learn), miscellaneous tools (WordNet, DBpedia, Linux Shell, Git).

Education

University of Toronto	2013 to 2019
Doctor of Philosophy in Industrial Engineering	

Toronto Metropolitan University (formerly Ryerson University)	2008 to 2011
Master's of Science in Computer Science	

Toronto Metropolitan University (formerly Ryerson University)	2004 to 2008
Bachelor of Science in Computer Science	

Research Funding

Public Health Agency of Canada (PHAC) (3 years; of a \$3mm project)	\$ 55,000/y	CAD	2024 - 2026
Mitacs Accelerate Postdoctoral Fellowship Award (1 year)	\$ 25,000/y	CAD	2021 - 2022
Mitacs Accelerate Postdoctoral Fellowship Award (2 years)	\$ 25,000/y	CAD	2019 - 2021
Wondeur AI, Postdoctoral Fellowship Award (2 years)	\$ 75,000/y	CAD	2019 - 2021

Awards / Grants / Achievements

Winner: Best Poster: Post-Doctoral Fellow Poster Competition			2020
Natural Sciences and Engineering Category, Lakehead University			
MIE Postdoctoral Fellowship Award (4 months)	\$	23,000 CAD	2019
ACM SIGSIM PADS Travel Grant Award	\$	1,000 USD	2017 - 2018
MIE Doctoral Completion Award	\$	9,500 CAD	2017 - 2018
Research Assistantship, U of Toronto, Industrial Engineering	\$	13,000/y CAD	2013 - 2018
U of Toronto Fellowship, Mechanical and Industrial Engineering	\$	9,500/y CAD	2013 - 2017
Travel Grant, AAAI Conference 2017	\$	450 CAD	2017
Ontario Centres of Excellence First Job Initiative	\$	37,500 CAD	2010 - 2011
Graduate Stipend, Ryerson University, Computer Science	\$	12,000/y CAD	2008 - 2010
Travel Grant, International Semantic Web Conference 2009	\$	1,155 USD	2009
Travel Grant, GECCO 2009	\$	750 CAD	2009
Apple Worldwide Developer's Conference Student Scholarship	\$	1,633 CAD	2008
TD Bank Continuing Education Assistance Award	\$	1,314 CAD	2004
TD Bank Continuing Education Assistance Award	\$	1,797 CAD	2002

Graduate Research Experience

PhD Thesis: Artificial Intelligence Planning Techniques for Emulating Agents with Application in Social Services

Mechanical and Industrial Engineering, University of Toronto **2013 to 2019**

Under the supervision of Professors Mark S. Fox and Michael Grüninger, my goal was to apply data-driven artificial intelligence (AI) and engineering methods to emulate behavior and predict the progress of social service clients in specific intervention programs. The first objective is to create a reasoning framework that captures the behavior of individuals who are outside of social norms or do not fit a standard "rational agent" model. This framework combines the areas of AI planning, cognitive science, and social sciences. The second objective was to create a client agent and a simulation environment that takes a holistic view of evaluating social services. This model pays special attention to life experiences, decision trajectories, and the environment of individual clients, rather than a transactional measure of service provisioning. A new ontology of social service needs captures the needs and motivating factors expressed by clients, mapped to appropriate services and resources. Empirical validation relies on the evaluation of personal and structural constraints faced by clients, including cognitive limitations, lack of community support, and housing stability. I have partnered with the Calgary Homeless Foundation, which has provided me with a data set that captures client progress in their Housing First program.

PhD Thesis (pdf): <http://hdl.handle.net/1807/97017>

The research committee included: Professor Marion Bogo, Professor of Social Work, and previously the Acting Dean, Associate Dean, Practicum Coordinator, and inaugural Sandra Rotman Chair in Social Work at the Faculty of Social Work, University of Toronto; as well as Dr. Vicky Stergiopoulos, Associate Professor in the Faculty of Medicine, Director of the Division of Adult Psychiatry and Health Systems in the Department of Psychiatry at the University of Toronto, and Physician-in-Chief and Clinician Scientist at Centre for Addiction and Mental Health.

Master Thesis: Using Decision Trees for Inductively Driven Semantic Integration and Ontology Matching

Computer Science, Ryerson University, Toronto **2008 to 2011**

Under the supervision of Professors Alireza Sadeghian and Mikhail Soutchanski, my thesis focused on generating decision trees for ontology concepts and applying them to Ontology Matching. It is often the

case that different ontologists, experts, and organizations create the vast majority of ontologies for internal use or for use in a narrow context. At the same time, their domains frequently overlap in a wider context. To assist in the reuse of ontologies, this thesis proposed a bottom-up technique for creating concept anchors that are used for ontology matching. The matching process is based on inductively derived decision tree rules. The matching algorithm identifies matching ontology concepts with an associated database used to derive the decision trees. This thesis also introduces several algorithm evolution measures and presents a set of use cases that demonstrate the strengths and weaknesses of the matching process. The technologies used include Protégé, OWL API, HermiT, WEKA and WordNet.

MSc Thesis (pdf): <http://bit.ly/BartG-MSc-Thesis>

Publications

1. Knowles, P., **Gajderowicz, B.** & Dugas, K. (2024) “Data-Centric Design: Introducing an Informatics Domain Model and Core Data Ontology for Computational Systems,” In *Proceedings of the 10th International Conference on Data Mining & Knowledge Management* (Vol. 14, No. 17), Conference Proceedings.
2. Rosu, D., **Gajderowicz, B.**, & Fox M.S. (2023). Representing Goals, Needs and Outcomes in Social Work. In *F. Toyoshima, M. Katsumi, E. Sanfilippo (Eds.), The Ninth Joint Ontology Workshops (JOWO 23)*, CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3637/paper15.pdf>
3. Fisher, A., **Gajderowicz, B.**, Latimer, E., Aubry, T., & Mago, V. (2022). BEAUT: An Explainable Deep Learning Model for Agent-Based Populations With Poor Data, *Knowledge-Based Systems*, 248. <https://doi.org/10.1016/j.knosys.2022.108836>
4. **Gajderowicz, B.**, Rosu, D., & Fox, M. S. (2022). Compass Event, Client, and Service Ontology: A Design Pattern for Social Services. In *T. P. Sales & M. M. Hedblom (Eds.), The Eighth Joint Ontology Workshops (JOWO 22)*, CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3249/paper2-OSS.pdf>
5. Rosu, D., Fox, M. S., & **Gajderowicz, B.** (2022). Compass Needs Ontology: A Design Pattern for Representing Needs in Social Work. In *T. P. Sales & M. M. Hedblom (Eds.), The Eighth Joint Ontology Workshops (JOWO 22)*, CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3249/paper1-OSS.pdf>
6. Fox, M. S., **Gajderowicz, B.**, Rosu, D., Turner, A., Lyu, L., & Lyu, D. (2022). An Ontological Approach to Analysing Social Service Provisioning, In *2022 IEEE International Smart Cities Conference (ISC2)*, 1–7. <https://doi.org/10.1109/ISC255366.2022.9922132>
7. Fox, M.S., Chowdhury, A., Zhang, J., **Gajderowicz, B.**, Abdulai, T., and Rosu, D. (2020). CAFO: The Common Approach Foundation Ontology, *Technical Report, Centre for Social Services Engineering, University of Toronto*. Available at csse.utoronto.ca
8. Fox, M.S., Chowdhury, A., Zhang, J., **Gajderowicz, B.**, Abdulai, T., Ruff, K., and Rosu, D. (2020). CACO: The Common Approach Core Ontology for Modeling Impact Models, *Technical Report, Centre for Social Services Engineering, University of Toronto*. Available at csse.utoronto.ca.
9. Fox, M.S., **Gajderowicz, B.**, and Ruff, R. (2020). Common Approach Indicator Vocabulary, *Technical Report, Centre for Social Services Engineering, University of Toronto*. Available at csse.utoronto.ca.
10. **Gajderowicz, B.** (2019). Artificial Intelligence Planning Techniques for Emulating Agents with Application in Social Services (Doctoral dissertation). Available at <http://hdl.handle.net/1807/97017>
11. **Gajderowicz, B.**, Fox, M.S., & Grüninger, M. (2018). The role of goal ranking and mood-based utility in dynamic replanning strategies. *Journal of Advances in Cognitive Systems*, (8), 211–230. Available at <http://cogsys.org/journal/volume6/article-6-14.pdf>

12. **Gajderowicz, B.**, Fox, M. S., & Grüninger, M. (2018). Ontology of social service needs: Perspective of a cognitive agent. In *Proceedings of the 2018 Joint Ontology Workshops, Cognition And Ontologies + Explainable AI* (pp. 1–12). Cape Town. Available at http://ceur-ws.org/Vol-2205/paper13_caos2.pdf
13. **Gajderowicz, B.**, Fox, M. S., & Grüninger, M. (2017). General model of human motivation and goal ranking. In *Proceedings of the 2017 AAAI Fall Symposium Series on a Standard Model of the Mind*, Arlington, VA: AAAI Press. Available at <https://cdn.aaai.org/ocs/15992/15992-69906-1-PB.pdf>
14. **Gajderowicz, B.**, Fox, M. S., & Grüninger, M. (2017). Requirements for emulating homeless client behaviour. In *the Proceedings of the AAAI Workshop on Artificial Intelligence for Operations Research and Social Good* (p. 7). San Francisco, CA: AAAI Press. Available at <http://eil.mie.utoronto.ca/wp-content/uploads/2015/06/BartG-socialgood-ws.pdf>
15. **Gajderowicz, B.**, Fox, M. S., Grüninger, M. (2014). Requirements for an Ontological Foundation for Modelling Social Service Chains. In *the Proceedings of the 2014 Industrial and Systems Engineering Research Conference*, Y. Guan, J. Liao (eds.), Montréal, Quebec. Available at <https://www.proquest.com/docview/1622308598>
16. **Gajderowicz, B.** (2011). Using decision trees for inductively driven semantic integration and ontology matching, (Master's thesis), Ryerson University, 350 Victoria Street, Toronto, Ontario, Canada. Available at https://rshare.library.torontomu.ca/articles/thesis/Using_decision_trees_for_inductively_driven_semantic_integration_and_ontology_matching/14656047
17. **Gajderowicz, B.**, Sadeghian, A. (2009). Ontology Granulation Through Inductive Decision Trees. In *Proceedings of the 4th International Semantic Web Conference Workshop on Uncertainty Reasoning for the Semantic Web*, Washington D.C, USA, pp. 39-50. Available at <https://ceur-ws.org/Vol-527/paper4.pdf>
18. **Gajderowicz, B.**, Sadeghian, A, Soutchanski, M. (2008). Ontology Enhancement Through Inductive Decision Trees. In *Uncertainty and Reasoning for the Semantic Web II*, da Costa, P.C.G., d'Amato, C., Fanizzi, N., Laskey, K.B., Laskey, K.J., Lukasiewicz, T., Nickles, M., Pool, M. (eds.), ISWC International Workshop, URSW 2008-2010 Revised Selected and Invited Papers. LNCS (LNAI). Springer, Heidelberg. Available at https://link.springer.com/chapter/10.1007/978-3-642-35975-0_14
19. Rahnama, H., Madni, A.M., Sadeghian, A., Mawson, C., **Gajderowicz, B.** (2008). Adaptive context for generic pattern matching in ad hoc social networks. In *Proceedings of the 2008 IEEE 3rd International Symposium on Communications, Control and Signal Processing, ISCCSP 2008*, art. no. 4537195, pp. 73-78 (2008). Available at https://link.springer.com/chapter/10.1007/978-3-642-35975-0_14

Under Review / In Progress

1. Fox, M. S; **Gajderowicz, B.**; and Dishu, L., “A Capability Maturity Model for Urban Dataset Metadata,” In *International Journal of Digital Curation* (under review), Accessible at: <http://doi.org/10.48550/arXiv.2402.05211>
2. **Gajderowicz, B.**, Fox, M.S. and Gao, Y., (in progress) “Towards an Ontology of Traceable Impact Management in the Food Supply Chain”, Accessible at: <https://doi.org/10.48550/arXiv.2501.05486>

Posters

1. Mark Fox, **Bart Gajderowicz**, and Dishu Lyu: “Canadian Urban Data Catalogue,” *International Association for Social Science Information Service and Technology (IASSIT)*, May 30, 2024.

2. **Gajderowicz, B.**, Fox, M.S., and Grüninger, M. (2018). "Limitations of human-centric decision making: An observer's perspective," In *Poster Proceedings of the 6th Conference on Advances in Cognitive Systems* (pp. 1–12). Stanford, CA: Cognitive Systems Foundation. Available at <http://www.cogsys.org/posters/2018/poster-2018-7.pdf>
3. **Gajderowicz, B.**, Sadeghian, A., and dos Santos, M. (2009). Expectation Maximization Enhancement with Evolution Strategy for Stochastic Ontology Mapping. In *Proceedings of the 11th Annual Conference on Genetic and Evolutionary Computation*. ACM, New York, NY, pp. 1847-1848. Available at <https://dl.acm.org/doi/abs/10.1145/1569901.1570197>

Patents and Standards

1. Fox, M.S., **Gajderowicz, B.**, and Lyu, Dishu, "Metadata Requirements for Urban Datasets", In *Digital Governance Standards Institute Technical Committee 6 (TC 6) on Connected Cities*, CAN/DGSI 121:202X (in public review)
2. **Gajderowicz, B.**, and Vijay, M. Intent elicitation in dynamic and heterogeneous networks with imperfect information. November 10 2022. US Patent App. 17/734,929 (pending), Available at <https://patents.google.com/patent/US20220358602A1/en>

Invited Talks and Panels

1. **Speaker: Bart Gajderowicz**, Daniela Rosu, Mark S Fox: "Enhancing Social Work with Evidence-driven Artificial Intelligence: Benefits and Concerns," *Evidence into Practice Special Interest Group (EIPSIG)*, as part of *European Social Work Research Association (ESWRA)*, October 29, 2024.
2. **Co-Author:** Daniela Rosu, **Bart Gajderowicz**, Mark S Fox: "Empowering Social Workers: Harnessing Knowledge and Data to Improve Client Outcomes," The International Journal on Homelessness Conference, Santiago, Chile, January 20, 2025
3. **Speaker: Bart Gajderowicz**, Mark S Fox, and Yongchao Gao: "Towards an Ontology of Traceable Impact Management in the Agriculture Food Chain," *Foundations for Food and Food Systems Ontologies Workshop*, May 21, 2024.
4. **Speaker: Bart Gajderowicz**, Mark S. Fox, and Dishu Lyu: "Dataset Metadata Maturity Model", *International Association for Social Science Information Service and Technology (IASSIT)*, May 31, 2024.
5. **Speaker: Bart Gajderowicz** and Mark S. Fox: "SMILE: An Explainable Artificial Intelligence System for Extracting Social Service Impact Models from Text," *13th European Conference for Social Work Research*, April 19, 2024.
6. **Penalist:** "Integrative knowledge, ontology, and data for ecosystem and policy that ensure lifelong health, wealth and wellbeing and multi-level resilience," *Ecosystem and Policy: Faculty of Management, Precision Convergence-Retailing*, McGill University. Panelists: Andrea Borghini, **Bart Gajderowicz**, Damien Dooley, John Keogh, February 2024.
7. **Speaker: Bart Gajderowicz:** "Towards an Ontology of Traceable Impact Management in the Agriculture Food Chain." *Series: The Convergent Innovation Webinar Series: Inventing "One-World" Solutions for Lifelong Wellness and Sustainable Economic Growth*. Panelists: Jurij Skornik, Troy Ruemping, Rob Warren, Horia Bradau, Naomi Kay Fukagawa. December 6, 2023
3. **Panelist:** Andrea Borghini (speaker): "Bots in the Kitchen: Philosophical Take on the Digital Food Transformation." *Series: The Convergent Innovation Webinar Series: Inventing "One-World" Solutions for Lifelong Wellness and Sustainable Economic Growth*. Panelists: Brian King, Catherine Paquet, **Bart Gajderowicz**, Ebenezer Miezah Kwofie. September 23, 2024.
4. **Speaker: Bart Gajderowicz.** "Survey of University of Toronto Ontology of Impact," *Government Ontology workshop, Basic Formal Ontology Summit Meeting*. May 23, 2023

5. **Speaker:** Matt Parker, **Bart Gajderowicz**, Lester Lym, Daniela Rosu, Alina Turner & Mark S. Fox , “Data Consolidation: How we Solved it with Kompas Ontology and Data,” *6th Annual Data That Makes a Difference Conference*, Calgary, Alberta, Canada, May 17th, 2023
6. **Seminar:** **Bart Gajderowicz**. ”Neural-symbolic artificial intelligence: state-of-the-art, what’s missing, and next steps,” *Department of Computer Science Graduate Seminar 2022*, Lakehead University, Thunder Bay, Ontario, Canada, September 23, 2022.
7. **Panelist:** “Research that is outside of the box,” *Research and Innovation Week*. Panelists: Salimur Choudhury, Martha Dowsley, **Bart Gajderowicz**, Lindsay Galway, Lana Ray, Pauline Sameshima, Lakehead University, Thunder Bay, Ontario, Canada. February 28, 2020
8. **Speaker:** **Bart Gajderowicz**, Mark S. Fox, and Michael Grüninger. *Artificial Intelligence Planning Techniques for Emulating Agents*: ACM SIGSIM PhD Colloquium, Chicago, Illinois, USA, June 4, 2019 (Awarded an ACM Travel Grant)
9. **Speaker:** **Bart Gajderowicz**, Mark S. Fox, and Michael Grüninger. *Requirements for Data-Driven Social Service Policy Evaluation*. Third Annual Data Sharing Initiative Calgary, Alberta, Canada, May 28th, 2018
10. **Panelist:** *Ethics in Artificial Intelligence*. Panelists: Dwija Patel, Matt Kantor, **Bart Gajderowicz**, and Myles Harrison. Toronto, Ontario, Canada, July 13, 2017
11. **Panelist:** *The Future of Semantic Web and its Applications*, Panelists: Mark van Berkel, **Bart Gajderowicz**, and James Leigh; Toronto, Ontario, Canada, April 4, 2017
12. **Speaker:** **Bart Gajderowicz**, Mark S. Fox, and Michael Grüninger. *Artificial Intelligence Planning Techniques for Emulating Agents, with Application in Social Services*. Machine Intelligence Toronto. Toronto, Ontario, Canada, March 9, 2017.

Service and Committees

Technical Committee: Common Approach: Common Impact Data Standard	2023 - 2025
Workshop Chair: Workshop on Ontologies for Services and Society (OSS)	2022 - 2025
Program Committee: Formal Ontology in Information Systems (FOIS)	2023 - 2025
Program Committee: Applied Ontology Journal	2024 - 2025
Program Committee: International Conference on Biological and Biomedical Ontology (ICBO)	2024
Program Committee: International Association for Social Science Information Service and Technology (IASSIST) Quarterly	2024
Program Committee: International Workshop on Narrative Extraction from Texts (Text2Story)	2023 - 2024
Program Committee: Semantic Web Journal (SWJ)	2022
Steering Committee: EAI International Conference on Social Data and AI (SDAI)	2019
Steering Committee: MIE Graduate Student Research Symposium	2015 - 2016
Review Committee: Computational and Mathematical Organization Theory	2019 - 2021
Steering Committee: Undergraduate Engineering Research Day, U of Toronto	2015 - 2017, 2020

Professional Research Experience

Senior Research Associate, University of Toronto

Urban Data Research Centre / Centre for Social Services Engineering October 2023 to current

Under the supervision of Professor Mark Fox, I am the Director of the SeMantIc roLe Extraction project (SMILE), a hybrid AI architecture that focuses on explainable natural language understanding AI models for extracting Impact Model information from unstructured text. My involvement includes designing and developing SMILE's blackboard architecture, knowledge sources and knowledge graph database, and supervising two to three students per semester. I am also the supervisor of two to three students each semester working on an NLP-to-SPARQL interface for knowledge graph databases. I also manage the Canadian Urban Data Catalogue project (CUDC), where my team develops tools for cataloging datasets about urban centres worldwide. In this role, I also supervise two to three students each semester. As a co-

author, I actively contribute to the Common Impact Data Standard (CIDS), an ontological basis for extracting Impact Model information, and serve on the Common Approach Technical Committee, advising on the technical direction of CIDS. I am the lead on the adaPtive grAph Repository pipeLine for dynAmic kNoledge sourCEs (PARLANCE) project, which involves developing data translation and consolidation logic, enabling the Compass platform to merge diverse data sources into a single representation stored in a knowledge graph. As part of my outreach efforts, I have organized workshops for academic and industry audiences and written multiple funding proposals.

Postdoctoral Fellow, University of Toronto

Data Modelling and AI Planning for Learning Social Service Needs/Satisfier Associations.

Centre for Social Services Engineering

August 2021 to September 2023

Under the supervision of Professor Mark Fox, I was a postdoctoral fellow at the Centre for Social Service Engineering. The goal of this research is to combine applied ontologies and machine learning to automate the construction of individual service plans within the domain of social services based on the theory of needs/satisfiers; a sequence of service interventions whose outcomes address the client's needs. My task was to create an entity and semantic role extraction model to convert unstructured text to a knowledge graph and generate a set of plans based on those services. Here, we created a set of ETL pipelines to transform various data sources about service providers, clients, and funders within the Compass project. Next, we created a hybrid (symbolic and sub-symbolic) model that extracts related knowledge using an ontology-driven search. The result of this research is both an ontology and methodology for dynamic needs/satisfier planning that can be incorporated into the Compass platform and made available throughout Canada. I served as a liaison between our data partner on the Compass project and our research assistants and engineers. I served as a project manager to ensure deadlines were met and resources utilized fully. I also performed outreach duties and organised workshops.

Postdoctoral Fellow, Lakehead University, Wondeur AI

Social simulation of strategic behaviour and contextual factors in the global art market.

Lakehead University / Wondeur AI

August 2019 to July 2021

Under the supervision of Dr Vijay Mago, I was a postdoctoral fellow at Lakehead University and a senior research scientist at Wondeur AI, our industry partner. This project is an extension of social simulation models developed during my PhD thesis. The models are used to combine machine learning and game-theoretical models with social simulation to capture the behavior of a market that is strongly dependent on subjective metrics, social norms, and market trends not sufficiently represented in available data. In my role as a senior research scientist at Wondeur AI, I was responsible for developing the simulation research program and the incorporation of social indicators. In cooperation with fellow members of the Wondeur AI senior research team, I am devising a publication strategy and mentoring junior machine learning engineers. In my role as a postdoctoral fellow at Lakehead University, I am assisting in supervising graduate students at DataLab, Dr. Mago's research lab in the Computer Science department. DataLab focuses on applied research in real-world social and economic problems. This research utilizes machine learning, natural language processing, simulation, and machine learning. The methods and tools we develop are used to collect and analyze information from various sources and communicate meaningful results to stakeholders and funder.

Postdoctoral Fellow, MIE and TCS

Tata Consultancy Service (TCS), Behavioural Business and Social Sciences

Mechanical and Industrial Engineering, University of Toronto

April to August, 2019

The Human Centric Systems Research Group of Tata Consultancy Services (TCS) R&I has been working on methods to create grounded, fine-grained agent models grounded in research in behavioral science. This work overlaps with the models I developed during my PhD, namely the fine-grained agent model of the goal-directed decision-making of economically and socially disadvantaged citizens in an urban

environment. The objective of the research project was to explore hybrid models of citizens' decision-making. My responsibilities included investigating planning algorithms to extend the originally myopic and resolute agent decision-making models. The models focused on heuristics planning, case-based reasoning, goal reasoning, just-in-time hierarchical planning, and the wisdom hierarchy.

Research Assistant for Prof. Mark S. Fox, CSSE

Mechanical and Industrial Engineering, University of Toronto

September 2013 to 2019

As a research assistant under the supervision of Professor Mark S. Fox, I was responsible for various initiatives at the Centre for Social Services Engineering (CSSE). I was a member of the Social Services Simulator project at the CSSE. I was responsible for defining the project's objectives, researching, and selecting possible solutions, and performing outreach activities amongst related groups, including engineering, government, and social science. I was responsible for finding organizations to partner with and attending relevant events. These partnerships include data sharing, collaboration on existing and upcoming intervention programs, data analysis and validation, program evaluation, and process analysis. I also advised graduate students and post-doctoral fellows at the CSSE on their projects, sharing my expertise on modelling, analysis, as well as technical direction and outreach initiatives.

Support Worker - (Research project domain knowledge and expertise gathering)

The Scott Mission, Toronto, Ontario

February to July, 2015

This position was part of my research at the CSSE into the shelter system in the city of Toronto, Canada. As an overnight support worker, I was responsible for attending to the clients' needs, ensuring their safety, and recording various metrics during my direct interaction with them. I observed and interacted with clients, interviewed front line workers and program directors to understand the unique circumstances and needs of clients.

Senior Research Associate – Ci2 and UPCL

Computer Science Department, Ryerson University, Toronto, Ontario

2007 to 2010

As a Senior RA at the Computational Intelligence Initiative Lab (Ci2) and founding member of the Ubiquitous and Pervasive Computing Lab (UPCL) at Ryerson University, I reported directly to the lab coordinator. Duties included mentoring junior RAs in developing research projects. I have been responsible for designing and building various systems, ranging from intelligent matching algorithms (resulting in a patent), remote robot controls, context-based content management systems, and distributed media capturing applications. I was also responsible for developing and presenting research projects to industry and academic partners. A detailed list is available upon request.

Mentoring and Supervision Activities

I have mentored and assisted in the supervision of PhD, Master's, and undergraduate students on research projects, focusing on developing their research and software engineering skills. I have advised on publications strategies and venues, and proofread numerous manuscripts, advising one student on a successful PhD grant application. Worked closely with two students on their academic and software development and mentored/supervised over 25 students on their work-study research projects.

PhD Students

Ranfateh Chattha (2024): PhD Student, Political Science, University of Toronto. Supervised on development of annotation vocabulary for social service and machine learning data corpus (2024).

Liwei Liao (2023-2024): PhD Student, Educational Leadership and Policy, University of Toronto. As a work-study position, I supervised Liwei on data annotation for an NLP project. Liwei was exposed to the CIDS impact model ontology, which was used to annotate unstructured text for testing and training purposes (2023-2024).

Yunhong Tian (2023-2024): PhD Student, Civil Engineering, University of Toronto. As a work-study position, I supervised Yunhong as an Urban Data Curation, metadata and taxonomy creation, geospatial and social service dataset analysis.

Gaurav Rao (2020): PhD Student, Applied Sciences, Saint Mary's University. Assisted with a successful Mitacs grant application for PhD Research (2020).

Master's Students

Jianyue Bai (2024): MEd in Developmental Psychology and Education, University of Toronto. Supervised on urban data curation, metadata modeling and taxonomy creation.

Hussein Ismail (2024): Bachelor of Applied Science, Industrial Engineering, University of Toronto. Supervised on urban data curation, metadata modeling and taxonomy creation.

Girirah Heda (2022-2023): MSc Student in Master's of Information, School of Information, University of Toronto. Supervised on urban data curation, metadata modeling, taxonomy creation, data analysis, and software engineering.

Marina Silic (2021): MSc Student, Medical Biophysics The University of Toronto. As a work-study position, I supervised Marian on hypothesis testing and ETL pipeline on the hybrid NLP project.

Pedram Khoshnevis (2019): assisted in the supervision of Computer Science Master's Thesis, Lakehead University. Defended in April 2021.

Dhivya Chandrasekaran (2019): MSc Student, Computer Science, Lakehead University. Assisted with editing numerous manuscripts. Currently a PhD student in Computer Science, Saint Saint Mary's University.

Andrew Fisher (2019-2020): MSc Student, Computer Science, Lakehead University. Assisted in the supervision of. Master's thesis. Defended in December, 2020. In 2024, graduated with a PhD from the Computer Science Department, Saint Mary's University. Currently a Postdoctoral Fellow at York University, Toronto, Canada.

Undergraduate Students

Kaviraj Gaire (2024): BSc Student, Computer Science, University of Toronto. Supervised Kaviarj during the development of a knowledge graph reasoning system for extracting impact model information from unstructured text.

Lian Hosseinnia (2024): Bsc Student, global Health and Bioethics, University of Toronto. Supervised on development of annotation vocabulary for social service and machine learning data corpus.

Janel Gilani (2023-2024): BSc Student, Computer Science Specialist & Statistics Minor, University of Toronto. Supervised Janel on NLP Validation methods using taxonomy, graph theory, and TF-IDF encoding. Utilized GridSearch for aligning various taxonomies with

Kasandra Tworzyanski (2024): Physical & Environmental Geography with minors in GIS and Statistics, University of Toronto. Supervised on urban data curation, metadata modeling and taxonomy creation.

Yifan Liu (2023-2024): BSc Student, Computer Science, University of Toronto. Supervised Yifan on Natural Language Models and software engineering methodology, including unit testing and architecture design.

Silin Lyu (2023-2024): BSc Student, Economics and Statistics, University of Toronto. Supervised on NLP model testing, software engineering methodology and unit testing. As an urban data curator, supervised Silin on metadata and taxonomy creation, and ETL pipeline software engineering.

Rudraksh Monga (2023-2024): BSc. Student, Computer Science; Minor: Mathematics and Statistics, University of Toronto. Supervised Rudraksh on LLM training and transfer learning methods, as well as ML optimization with GridSearch.

Aaron Lio (2023-2024): BASc - Engineering Science, University of Toronto. Supervised on Prompt Engineering for Named Entity Extraction using ChaptGPT for extracting impact model concepts.

James Hariady Widjaja (2023): BSc Student, Computer Science & Economics, University of Toronto. Working with a linguist, I supervised James on NLP models, specifically evaluation and correction of dependency trees.

Leonid Nediak (2023): BSc Student, Mathematics and Computer Science Specialisation, University of Toronto. Supervised Leonid on creating a explainable NLU benchmarking suite, with a focus on Named Entity Extraction, with complex confusion matrix architecture and metrics.

Sunnie Hu (2023): BA Student, Health Studies with Statistical Studies and Visual Sciences, University of Toronto. As a work-study position, I supervised Sunnie on data annotation for an NLP project. Sunnie was exposed to the CIDS impact model ontology, which was used to annotate unstructured text for testing and training purposes.

Chaerin Song (2022-2023): BScs Student, Statistics and Linguistics, University of Toronto. Supervised Chaerin in performing hypothesis testing and developing a hybrid AI (symbolic and sub-symbolic) model for semantic relation extraction. Chaerin developed a prompt-generating process for Question/Answer LLMs.

Akash Shetty (2019-2021): BSc Student, Honors Math and Physics, McMaster University. Supervised Akash as an AI Research Intern at Wondeur AI on community detection and dimensionality reduction methods.

Professionals

Dishyu Lyu (2021-2024): Research Assistant, Urban Data Centre, School of Cities, University of Toronto. Supervise Dishyu during the developing and testing for a data integration ETL pipelines. Performed ontology engineering for semantic integration portion of the pipeline. Supervised developing and testing the Canadian Urban Data Catalogue and the Dataset Metadata Maturity Model.

Sharanya Ramidi (2019-2021): Machine Learning Engineer at Wondeur AI. Managed and mentored Sharanya on NLP tasks, creating text extraction scripts and evaluation criteria. Taught computer programming principles for data analysis.

Suman Kumar (2019): Software Engineer, Tata Consulting Services. As part of my postdoctoral position with TCS, I worked closely with Suman (PI) and two junior machine learning engineers: supervised and taught AI planning methods, case-based reasoning, and agent-based simulation, 2019.

Various employees and co-op students (2011-2014): As a lead semantic engineer and the CTO, I interviewed, mentored and supervised a number of software engineers.

Highschool Students

Nilanthi Balendra, Susan Feng, and Nika Zolfaghari (2008): While at the Computer Science Department, Toronto Metropolitan University, I mentored three high-school interns as part of the Research Opportunity Programs in Engineering and Science and Women in Engineering initiative at the.

Teaching Experience

MIE1513 (451) Instructor – Mechanical and Industrial Engineering Department

University of Toronto, Toronto, Ontario

1 semester: 2015

MIE1513 Decision Support Systems: This course is co-taught as the undergraduate course MIE 451. During the Fall 2015 semester, my responsibilities include designing the syllabus, creating and modifying slides for the entire course, managing one teaching assistant, creating exams, designing projects, mentoring students, and grading assigned coursework. The course covers various techniques for information analysis, and knowledge-based problem-solving methods such as heuristic search, automated deduction, constraint satisfaction, and knowledge representation. Prolog is used as the main implementation language. Graduate students have a choice of a final project in Prolog or a combination of a Prolog and a research report.

CXCP685 Instructor – Information Technology Studies**Ryerson University, Toronto, Ontario****3 semesters: 2010 to 2011**

CXCP685 PHP/MySQL Web Design Fundamentals: Responsibilities included creating, administering, and grading lecture notes, assignments, tests, and a final exam. The PHP language is used to introduce programming constructs and web application principles. MySQL is used to introduce database basics. The course teaches students how to develop dynamic and secure web applications with a database back-end.

CPS125 Instructor – Computer Science Department**Ryerson University, Toronto, Ontario****2 semesters: 2009 to 2010**

CPS125 Digital Computation and Programming: Responsible for over 75 engineering students. The C programming language is used to introduce students to computer architecture, as well as programming principles and best practices. Topics include computer architecture, C syntax, programming constructs, file I/O, strings, arrays, and multidimensional matrices.

MIE457 Teaching Assistant - Mechanical and Industrial Engineering Department**University of Toronto, Toronto, Ontario****1 semester: 2017**

MIE457 Knowledge Modelling and Management: My responsibilities included creating and presenting tutorials every week on the topics covered. I was also one of two TAs responsible for administering laboratory lessons, assisting students with the technologies used, and answering questions. This course explores both the modelling of knowledge and its management within and among organizations. Knowledge modelling focuses on knowledge types and their semantic representation. It reviews emerging representations for knowledge on the World Wide Web (e.g. RDF and OWL). Emerging knowledge modelling and automated reasoning software were used in the laboratory.

MIE350 Teaching Assistant – Mechanical and Industrial Engineering Department**University of Toronto, Toronto, Ontario****3 semesters: 2013 to 2016**

MIE350 Design and Analysis of Information Systems: My responsibilities included creating and presenting slides focusing on Object-Oriented Programming and Web Applications, meetings with students regarding their projects, and administering quizzes. This course explores the software lifecycle, encompassing the planning, design, analysis, and implementation of software systems. Students learn techniques and methodologies for requirements engineering, data flow diagrams, process modelling, and UML; these techniques are applied in a course project.

MIE1512 Teaching Assistant - Mechanical and Industrial Engineering Department**University of Toronto, Toronto, Ontario****1 semester: 2016**

MIE1512 Data Analytics: My responsibilities included creating and marking all non-lecture material, including assignments and laboratories. I was responsible for learning multiple data analytics technologies, including Python and Scala Notebooks, IBM Bluemix, and experimental systems, including IBM Data Scientist Workbench. The course required me to evaluate each package, its strengths and weaknesses, and provide students with instructions on using these technologies for their laboratory assignments and projects. This course is a research seminar that focuses on recent developments in the area of Data Analytics and Big Data. This seminar provides an overview of data analytics concepts, approaches, and techniques, including distributed computations on massive datasets and frameworks for enabling large-scale parallel data processing on clusters of commodity servers. Emphasis is given to algorithmic techniques for analyzing Web Data. The project goal is to prepare publishable research contributions in the area of data analytics.

MIE253 Teaching Assistant – Mechanical and Industrial Engineering Department**University of Toronto, Toronto, Ontario****1 semester: 2016**

MIE253 Data modeling: As the head TA, I was responsible for managing seven TAs and organizing laboratory and assignment grading schedules. I was also responsible for giving lectures, as well as creating and marking assignments. This course provides an understanding of the principles and techniques of information modelling and data management, covering both relational theory and SQL database systems (DBMS), as well as entity-relation conceptual modelling. The course also familiarizes the student with analytical applications (OLAP) and provides an introduction to XML data management. The laboratory focuses on database application development using SQL DBMS, OLAP queries and data modelling.

MIE451 Teaching Assistant – Mechanical and Industrial Engineering Department**University of Toronto, Toronto, Ontario****1 semester: 2014**

MIE451 Decision Support Systems: Responsibilities included creating and presenting slides focusing on data mining, various decision support systems, Object Oriented Programming using Java, meetings with students regarding their projects, and creating and marking exams and projects.

CPS630 Teaching Assistant – Computer Science Department**Ryerson University, Toronto, Ontario****3 semesters: 2007 to 2010**

CPS630 Advanced Web Applications: TA duties included assisting students with course content and conducting tutorials; creating and marking assignments/tests; organizing guest lecturers; co-organizing a student project competition judged by both academic and industry individuals. Topics: Web 2.0 Principles, Ajax technology/frameworks, Google Maps/APIs, Web Services, Apache, Tomcat, JSP, PHP, Ruby on Rails, MySQL, JavaScript, Web media, Mobile APIs.

CPS109 Teaching Assistant – Computer Science Department**Ryerson University, Toronto, Ontario****1 semester: 2008**

CPS109 Computer Science I: TA duties included assisting students with course content and conducting tutorials; marking assignments, mid-terms and final exams; conducting labs. Topics included an introduction to programming techniques, concepts, control structures, and Object-Oriented programming, and graphics using the Java 1.5 programming language.

CPS209 Teaching Assistant – Computer Science Department**Ryerson University, Toronto, Ontario****1 semester: 2007**

CPS209 Computer Science II: TA duties included assisting students with course content and conduct tutorials; mark assignments, mid-terms, and final exams; conduct labs. Topics included inheritance and polymorphism, Swing, Multithreading, enumeration, Object Oriented Design.

Professional Experience**Senior Developer, Influitive Inc., Toronto, Ontario****January to August, 2013**

As a Senior Developer, I was responsible for integration of the Engagio (see below) architecture into the Influitive system workflow. I worked directly with the product management, customer support and engineering teams to develop and implement an integration strategy. I provided direction on designing solutions and developed modules for high scalability. I was also the first technical lead on a new internal “growth hacker” team, which was tasked with the design and implementation of short and highly iterative projects. A project’s success was rated using the A/B Test methodology, and successful projects resulted in formal specification requirements and an implementation schedule. I also lead several initiatives to incorporate machine learning into the Influitive workflow, providing business cases and technical solutions.

Co-Founder and Chief Technology Officer

Engagio Inc. (acquired by Influitive Inc.), Toronto, Ontario

2011 to 2013

As the CTO and founding member, I was responsible for designing and implementing the technical direction of Engagio and building the technical team. We worked in a highly agile development environment and adopted the “lean startup” methodology. We built an architecture capable of consuming over 1 million comments per day. I was responsible for taking Engagio from conception to most-viable-product through to a production-ready application. I worked directly with our CEO, marketing department, business analysts, engineers, and designers to develop the tools required to analyze system performance and collect statistics. Our engineering goal was a highly scalable and efficient system for growth and agility.

Software Engineer, Semantic Technologies, Equentia Inc., Toronto, Ontario

2010 to 2012

As a Software Engineer at Equentia Inc., I was responsible for incorporating semantic technologies and natural language processing into the existing workflow of our system, which processes over 100,000 articles per day. The goal was to improve the quality of text classification and topic extraction on these articles. I was also responsible for building tools to manage interrelated taxonomies on various topics, which ensured their soundness throughout the maintenance process. I incorporated YAGO and DBpedia to enrich the taxonomies with external corpora. I was also responsible for reviewing open-source packages and incorporating them into the Equentia system. For packages that resulted from academic research, I was also responsible for reviewing any associated publications.

TD Canada Trust, Solutions Developer – Enterprise Technology Solutions

2001 to 2007

Objective was to work with Agile and Project Life Cycle paradigms to develop financial systems. My responsibilities were project management, system analysis, design, development, testing, maintenance, and production support, as well as developing ad-hoc statistical analysis and data mining applications. The technologies used were XML, XPath, XSL, Ruby, Paradox, COBOL, SAS.

Web Application Consultant

NuLayer Inc.

February to April, 2010

Reporting to the company founders, the objective was to develop a frontend AJAX heavy application for Ruby on Rails workflow application, which produces printable PDF forms and debug an existing application. The technologies used were Ruby on Rails, Prototype, CSS, and Prince PDF library.

Celect.org

September 2009 to January 2010

Reporting to the Chief Development Officer, the objective was to develop a complete awards application, which allowed administrators to create awards, categories, and the actual online questionnaires to be filled out by judges and receive submissions from candidates. I was also responsible for consulting on the design and user experience of the site, implementing the designs in CSS, recommending a hosting service, setting up the entire Linux/Apache-based web server, and configuring a github.com repository. The system was designed to handle multiple organizations, with multiple awards and various levels of users. The AJAX-equipped questionnaire builder included different types of questions such as text, radio buttons, checkboxes, and file uploads, including specific file types for PDF and photos. The technologies used were Ruby, Ruby on Rails, Postgres, Prototype, CSS, RSpec, Capistrano, Linux, and Apache.

Mikemap.com – Independent Music and Event Management

October 2008 to April 2009

Objective was to assist with a startup, and act as their Chief Technology Officer. My duties were to design and implement a localization-based music portal for the promotion, torrent publishing, and business development of independent artists around the world. My main responsibilities included consulting on legal matters, determining technology and architectural direction, coordinating development efforts, and overseeing 3 developers and 3 designers. I performed lead development duties and worked with junior

programmers on implementing the site architecture and various site features. The technologies used were Ruby, Ruby on Rails, CSS, AJAX, MySQL, Google Maps API, YouTube, Vimeo.

Athletes Video - Social Network

April 2008 to November 2008

Reporting directly to the CEO, the objective was to design and develop a social network built around the Media Management System mentioned below, which manages athlete analysis videos, event media, and event websites. My responsibilities were to design and develop a social network for sports enthusiasts, of which a central component is the aforementioned management system. I was also responsible for implementing the CSS based on the designer's mockups. The system included internal messaging, email notifications, user profile management, assigning favorite videos, a photo gallery, and user comments for videos and photos. The site required detailed statistics on page views, as well as tracking the number of times videos were viewed, and looped. I was also responsible for creating the legal contract, gathering requirements, mentoring a junior developer, as well as proposing and configuring production and development web servers. The technologies involved were Ruby, Ruby on Rails, AJAX, Prototype, Flash, CSS, MySQL, Apache, Mongrel, Capistrano, and Subversion.

Athletes Video – Media Management System

February 2007 to October 2007

Reporting directly to the CEO, the objective was to design and develop a media management system which manages sporting events and associated media. My responsibilities were to design and develop a system which manages videos, photos, event sites, and users. The system has an online and batch process for creating content used by users, field organizers, and administrators. The system worked with multiple organizations, custom designs and branding, custom domains, and event subdomains. I was also responsible for creating the legal contract, gathering requirements, mentoring a junior developer, as well as proposing, configuring and maintaining production and development web servers. The technologies involved were Ruby, Ruby on Rails, AJAX, Prototype, CSS, MySQL, Apache, Mongrel, Capistrano, and Subversion.

Appointments

Workshop Chair: Workshop on Ontologies for Services and Society (OSS)	2022 - 2023
Vice President, AMIGAS, University of Toronto student group at MIE	2015 - 2016
My duties included: liaison with MIE department faculty and staff, advise and coordinate Academic Committee initiatives with Academic VPs; outreach program to IE members and student groups; advisor to the MIE 2016 Graduate Research Symposium Committee.	
IE Representative, AMIGAS, University of Toronto student group at MIE	2014 - 2015
My duties included representing and supporting Information Engineering (IE) initiatives.	
MIE 2015 Graduate Research Symposium Co-Chair.	
President, Computer Science Graduate Student Association (CSGSA)	2009 - 2010
CSC Graduate Student Representative, Computer Science Council	2009 - 2010

Professional Memberships

Association for the Advancement of Artificial Intelligence (AAAI)	2017 - present
International Association for Ontology and its Applications (IAOA)	2009 - present
Association for Computing Machinery Student Membership (ACM)	2009 - present
Institute of Electrical and Electronics Engineers (IEEE)	2009 - present

Hobbies

In my free time, I enjoy traveling, listening to jazz, practicing the trumpet, strength-training, jogging, and cycling.

References

Available upon request.